1 Lemmas on Error of Local Linear Estimator

1.1 Commonly Used Lemmas

Let's first state lemmas that I have seen everywhere but I want to make sure that my understanding is correct.

Lemma 1.1 (Multivariate Taylor's Theorem). *Let E be an open subset of $\mathbb{R}^n$, $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be k-th differentiable, that is if all its k-th partial derivatives are continuous. Then we have for any $x \in E$,*

$$f(y) = P_k(y) + R_k(y)|y - x|^k$$

where $P_k(y) = \sum_{|\alpha| \leq k} \frac{D^\alpha}{\alpha!} f(x)(y - x)^\alpha$ and $\lim_{y \to x} R_k(y) = 0$.

Lemma 1.2. *If A_n, e_n are matrices with random entries and $A_n \longrightarrow A$, $e_n = o_p(1)$, then $(A_n + e_n)^{-1} = A^{-1} + o_p(1)$*

1.2 Calibrate the First-Step Error

Let Y_{it} be the excess returns. We want to test

$$H_0 : E(Y_i \mid X_i) = X_i' \gamma \quad a.s. \quad \text{for some} \quad \gamma \in \Gamma \tag{1}$$

$$H_1 : E(Y_i \mid X_i) = m_1(X_i) \neq X_i' \gamma \quad \text{for all} \quad \gamma \tag{2}$$

where $X_{ik} = \operatorname{cov}(Y_i, F_k) / \operatorname{var}(F_k)$ and we assume the time-series dynamic is

$$Y_{it} = X_i' F_t + \varepsilon_{it} \tag{3}$$

where ε_{it} are assume to be indepepdent across t and independent of F_t.

Following Shanken, 1992, if H_0 is true and Equation 3 holds, let $m_0(x, \gamma) = x' \gamma$, because

$$E(Y_i) = X_i' E(F) \quad \bar{Y}_i = X_i' \bar{F} + \bar{\varepsilon}_i \tag{4}$$

$$E(Y_i) = m_0(X_i, \gamma) \quad \bar{Y}_i = \bar{m}_0(X_i, \gamma) + \bar{\varepsilon}_i \tag{5}$$

we have

$$\bar{m}_0(X_i, \gamma) = m_0(X_i, \gamma) + X_i'\left(\bar{F} - E(F)\right)$$

Let $\bar{M} = \begin{bmatrix} \bar{m}_0(X_1, \gamma) \\ \vdots \\ \bar{m}_0(X_n, \gamma) \end{bmatrix}$ and $\bar{\varepsilon} = \begin{bmatrix} \bar{\varepsilon}_1 \\ \vdots \\ \bar{\varepsilon}_n \end{bmatrix}$.

1.3 Local Linear Estimator

As I haven't found extension of the Generalized Likelihood Ratio test to the case with dependent data, I am going to use the method in Escanciano, Pardo-Fernández, and Keilegom, 2018 which still requires estimation of ε under the alternative. For any variable a_{it}, $\bar{a}_i = \frac{1}{T}\sum_t a_{it}$. In the first step we use OLS,

$$\hat{X}_i = (F'F)^{-1}F'Y_i \quad \text{so} \quad \hat{X} = Y_{(n\times T)}F\left(F'F\right)^{-1}$$

For $x \in \text{supp}(X) \subset \mathbb{R}^k$, let local linear estimator be $\hat{m}(x) = e_1^\top\left(\hat{X}_x'\hat{W}_x\hat{X}_x\right)^{-1}\hat{X}_x'\hat{W}_x\bar{Y}$ where $e_1^\top = (1, 0, \dots, 0)'$

$$\hat{X}_x = \begin{bmatrix} \mathbf{1} & \left(\hat{X} - \mathbf{1} \otimes x'\right) \end{bmatrix} \qquad \hat{W}_x = \text{diag}\left(K_h\left(\hat{X}_1 - x\right), \dots, K_h\left(\hat{X}_n - x\right)\right)$$

Let $\theta(x) = (m(x), Dm(x))'$.

Assumption 1. *We restrict the alternative by assuming $m(x) \in \mathcal{C}^2$. Let $K(\cdot)$ be a symmetric probability density function with compact support. $K_h(\cdot) = \frac{1}{h}K\left(\frac{\cdot}{h}\right)$.*

$$\begin{aligned}
\hat{\theta}(x) - \theta(x) &= \left(\hat{X}_x'\hat{W}_x\hat{X}_x\right)^{-1}\hat{X}_x'\hat{W}_x\bar{Y} \\
&= \hat{S}_x^{-1}\hat{X}_x'\hat{W}_x\left(\bar{M} + \bar{\varepsilon} - \hat{X}_x\theta(x)\right) \\
&= \hat{S}_x^{-1}\hat{X}_x'\hat{W}_x\left(M - X_x'\theta(x) + X(F_t - EF) + \left(\hat{X}_x - X_x\right)'\theta(x) + \bar{\varepsilon}\right)
\end{aligned}$$

where $\hat{S}_x = \hat{X}_x'\hat{W}_x\hat{X}_x$.

Lemma 1.3. *We have the following decompositions:*

1. $\hat{X}_x = X_x + R_1$ *where* $R_1 = \begin{bmatrix} \mathbf{0} & \hat{X} - X \end{bmatrix} = O_p\left(\frac{1}{\sqrt{T}}\right)$;

2. $\hat{W}_x = W_x + R_2$, *where* $R_2 = \operatorname{diag}\left\{\frac{1}{h^2}K'\left(\frac{X_1 - x}{h}\right)\left(\hat{X}_1 - X_1\right), \ldots, \frac{1}{h^2}K'\left(\frac{X_n - x}{h}\right)\left(\hat{X}_n - X_n\right)\right\}$.

3. $\hat{S}_x = S_x + o_{p,n}(1) + O_p\left(\frac{1}{\sqrt{T}}\right)$.

Proof. $\hat{X}_x = \begin{bmatrix} \mathbf{1}_n & \hat{X} - \mathbf{1}_n x' \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n & X - \mathbf{1}_n x' \end{bmatrix} + \begin{bmatrix} 0 & \hat{X} - X \end{bmatrix} = X_x + O_p\left(\frac{1}{\sqrt{T}}\right)$.

Assuming K_h has bounded support and bounded first-order derivative, then by Taylor expansion we have the estimate for $\hat{W}_x$.

□

Proposition 1.4 (MSE of Local Linear Estimator)**.**

1.4 Specification Test

We show that this fits into the framework of Escanciano, Pardo-Fernández, and Keilegom, 2018. We calculate the following quantities:

$$\hat{\varepsilon}_{j,0} = \frac{Y_j - m_0\left(X_j, \hat{\gamma}\right)}{\hat{\sigma}(X_j)} \quad \text{and} \quad \hat{\varepsilon}_j = \frac{Y_j - \hat{m}\left(X_j\right)}{\hat{\sigma}(X_j)}$$

References

Escanciano, Juan Carlos, Juan Carlos Pardo-Fernández, and Ingrid Van Keilegom (June 2018). "Asymptotic Distribution-Free Tests for Semiparametric Regressions with Dependent Data". In: *Annals of Statistics* 46.3, pp. 1167–1196. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/17-AOS1581. URL: https://projecteuclid.org/euclid.aos/1525313079 (visited on 06/17/2020).

Shanken, Jay (Jan. 1, 1992). "On the Estimation of Beta-Pricing Models". In: *The Review of Financial Studies* 5.1, pp. 1–33. ISSN: 0893-9454. DOI: 10.1093/rfs/5.1.1. URL: https://academic.oup.com/rfs/article/5/1/1/1599649 (visited on 02/22/2020).